
BerryBrain: Architecture and Experimental Evaluation of an Adaptive Knowledge Management System Based on Semantic Graphs, GraphRAG, and Human Feedback

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Abstract. Personal knowledge management systems preserve notes under user control, but they often rely on manual links and lexical retrieval. Retrieval-augmented generation improves access to stored knowledge, yet purely vector-based approaches may miss cross-document relationships and generate answers whose provenance is difficult to audit. This work presents *BerryBrain*, a self-hosted system that converts Markdown notes into a governed semantic graph, combines lexical, dense, and structural retrieval through GraphRAG, and incorporates provenance, confidence estimates, a configurable panel of language-model judges, and reversible human feedback. The study follows Design Science Research and evaluates version 1.4.8 through automated tests, paired retrieval ablation, a public benchmark, and performance, fault, security, accessibility, and maturity assessments. On a controlled 44-query fixture, graph-hybrid retrieval achieved Recall@10 of 1.00, compared with 0.50 for hybrid retrieval without graph expansion. On 20 multi-hop queries, the difference was 1.00 with an exploratory bootstrap interval of [1.00, 1.00]. This result validates the intended regression behavior, not external superiority. On BEIR SciFact, an independent BM25 implementation achieved Recall@10 of 0.7816, MRR of 0.6386, and nDCG@10 of 0.6644 over 300 queries. API statement coverage reached 82.19%, and Worker coverage reached 48.94%; 465 API tests, 55 Worker tests, and 56 end-to-end scenarios passed. On a Raspberry Pi 4, the API sustained 62.02 requests/s at 274.36 ms p95, while a synthetic 10,000-node graph completed loading in 6.35 s. The evaluation supports architectural coherence and regression behavior but does not yet validate human calibration of the judges, semantic generalization, or longitudinal learning from users. The work contributes an auditable architecture, an edit-and-delete lifecycle, and a reproducible protocol that distinguishes non-parametric adaptation from model-weight training.

Keywords: personal knowledge management, personal knowledge graph, GraphRAG, information retrieval, provenance, LLM as a judge, human feedback, self-hosted systems

1 Introduction

Personal notes record decisions, readings, hypotheses, and experience, but their value depends on the ability to retrieve not only documents but also relationships, contradictions, and knowledge gaps. Personal knowledge management (PKM) supports this process, while personal knowledge graphs (PKG) formalize entities and relationships controlled by an individual Balog and Kenter [2019]; Chakraborty and Sanyal [2023]; Skjæveland *et al.* [2024]. Recent systems use natural language processing to suggest connections between notes Fraga *et al.* [2023, 2024], but governance, provenance, update semantics, and evaluation remain open problems.

Retrieval-augmented generation (RAG) conditions language models on external evidence Lewis *et al.* [2020]. Dense retrievers are effective at semantic similarity Karpukhin *et al.* [2020], but fragment similarity alone does not represent paths across sources. GraphRAG adds relational structure, communities, and corpus-level summaries Edge *et al.* [2024]; HippoRAG models associative memory Gutiérrez *et al.* [2024]; and LightRAG proposes incremental graph-vector retrieval Guo *et al.* [2024]. These advances do not remove the need to track the origin of every claim, invalidate derivatives when sources change, or measure the

quality of automated evaluators.

The research problem is: *to what extent does an architecture that combines a semantic graph, hybrid retrieval, multi-agent validation, and explicit feedback improve relevance, traceability, and knowledge coherence over configurations without graph expansion, and what costs and limitations does this combination introduce?*

The general objective is to design, implement, and experimentally evaluate an adaptive architecture that converts personal notes into structured, queryable, and traceable knowledge. The specific objectives are:

1. extract concepts, entities, topics, evidence, and relationships from notes;
2. build a graph with governed classes, predicates, provenance, and lifecycle constraints;
3. integrate lexical, vector, and structural retrieval in a controlled GraphRAG pipeline;
4. validate and enrich artifacts through rules, agents, and configurable judges;
5. estimate confidence from observable signals without allowing manual score editing;
6. incorporate explicit user decisions as auditable and reversible feedback; and
7. evaluate retrieval, coherence, performance, resilience, se-

curity, accessibility, and architectural maturity.

This work contributes: (i) a local-first architecture that separates canonical notes from derived artifacts; (ii) an operational ontology shared by extraction, visualization, and RAG; (iii) an invalidation protocol for edits and deletions; (iv) feedback-guided non-parametric adaptation; and (v) a reproducible evaluation that states its evidential limits. The system is not claimed to learn foundation-model weights. Learning here changes graph state, policies, preferences, and suppression decisions.

2 Background and Related Work

2.1 Personal knowledge management and graphs

PKGs organize facts and relationships around one user, with ownership, privacy, and personalization requirements that differ from enterprise knowledge graphs Balog and Kenter [2019]. Recent surveys identify graph population, interoperability, utilization, and evaluation as open issues Chakraborty and Sanyal [2023]; Skjæveland *et al.* [2024]. Fraga *et al.* [2024] demonstrate automatic PKM connections and provide the closest peer-reviewed comparator for note linking. *BerryBrain* extends this scope by treating admission, provenance, lifecycle, structural retrieval, and feedback as one contract.

2.2 RAG, GraphRAG, and multi-hop retrieval

RAG integrates retrieval and generation Lewis *et al.* [2020]. BM25 remains a strong lexical baseline across heterogeneous collections, as shown by BEIR Thakur *et al.* [2021]. Dense passage retrieval learns query and passage representations Karpukhin *et al.* [2020], while Reciprocal Rank Fusion combines ranked lists without requiring comparable score scales Cormack *et al.* [2009]. Multi-hop datasets such as HotpotQA and MuSiQue require evidence composition across sources Yang *et al.* [2018]; Trivedi *et al.* [2022].

GraphRAG constructs relationships and communities for local and global questions Edge *et al.* [2024]. HippoRAG uses graphs and personalized PageRank for associative retrieval Gutiérrez *et al.* [2024]. HippoRAG 2 deepens passage integration and frames external memory as non-parametric continual learning across factual, sense-making, and associative tasks Gutiérrez *et al.* [2025]. LightRAG separates low- and high-level retrieval and emphasizes incremental updates Guo *et al.* [2024]. This work does not claim universal superiority over these systems; it evaluates an integrated artifact and uses internal ablations to isolate structural expansion.

2.3 RAG evaluation and LLM as a judge

RAGAS measures faithfulness and relevance without always requiring textual references Es *et al.* [2024]; ARES combines synthetic data, human annotations, and statistical inference Saad-Falcon *et al.* [2024]. LLM evaluators may align with human preferences, but they are not oracles Liu *et al.* [2023]. Specialized evaluators Kim *et al.* [2024] and position bias

Shi *et al.* [2025] reinforce the need for calibration against independent human reviewers.

In *BerryBrain*, judges receive separate faithfulness, relevance, and contradiction roles. The active generation model is excluded from the committee by default. Failures remain in provenance but do not enter voting. This design reduces coupling without turning synthetic agreement into evidence of human validity.

2.4 Provenance, ontology, and confidence

RDF, OWL, SKOS, and PROV-O provide vocabularies for identity, semantics, and provenance W3C [2014, 2012, 2009]; Lebo *et al.* [2013]. *BerryBrain* uses a smaller domain-specific operational ontology with domain and range constraints. Runtime confidence is a bounded summary of observed support signals, not a calibrated probability of truth. This distinction follows calibration research Guo *et al.* [2017] and prevents model output from being interpreted as certainty.

Table 1. Positioning against related lines of work.

Line	Primary unit	Main strength	Gap addressed in <i>BerryBrain</i>
PKG/PKM	personal notes, entities, and links	ownership and organization	lifecycle, feedback, and integrated evaluation
Vector RAG	semantically similar fragments	semantic retrieval	multi-hop relations and structural provenance
GraphRAG	entities, edges, and communities	corpus-level sense-making	local-first governance and source invalidation
LLM as a judge	automated verdicts	scalable evaluation	configurable committee, failure records, and explicit calibration

3 Research Method

3.1 Research strategy

The study follows Design Science Research Methodology (DSRM), whose activities comprise problem identification, solution objectives, design and development, demonstration, evaluation, and communication Hevner *et al.* [2004]; Peffers *et al.* [2007]. The artifact is *BerryBrain*1.4.8. Its design principles are treated as knowledge contributions only to the extent supported by the reported evidence.

This is an exploratory engineering evaluation. It used no human participants, personal note content, or model training. Deterministic fixtures exercised internal contracts, while BEIR SciFact supplied a public external reference for BM25. The confirmatory phase includes an annotated corpus, blinded human reviewers, and a longitudinal study; no values were simulated for those pending studies.

3.2 Research questions

- RQ1:** Does graph expansion improve multi-hop retrieval without an observed factual loss?
- RQ2:** Do ontology, provenance, and judge controls block unsupported claims and stale evidence?
- RQ3:** Do creation, editing, and deletion preserve graph coherence?
- RQ4:** Are evidence-support estimates signal-derived and auditable, and are they calibrated against correctness?
- RQ5:** Is the judge committee regression-stable and aligned with human reviewers?
- RQ6:** What latency, memory, payload, queue, and frontend costs are introduced?
- RQ7:** Does the system improve human tasks over search and non-graph RAG?
- RQ8:** Does explicit feedback reduce recurrence of rejected artifacts?
- RQ9:** Does the system preserve state and recover from failures?

3.3 Comparative design

Four layers prevent invalid comparisons. Internal ablation keeps corpus, qrels, order, context limit, and cache policy fixed while varying lexical, dense, hybrid, and graph expansion. Independent technical baselines implement BM25, dense retrieval, hybrid retrieval, and vanilla RAG on the

same corpus. Historical regression compares frozen revisions. User studies compare complete workflows and require an approved ethics protocol. Only the first layer and an external BM25 implementation check were executed in this stage.

Retrieval metrics were Recall@10, MRR, and nDCG@10. Generation and graph outcomes included citation support, unsupported claims, ontology violations, stale residue, and idempotence. Performance outcomes included median, p95, p99, throughput, peak memory, and payload. The exploratory paired interval used percentile bootstrap with 2,000 resamples and seed 20260812. Confirmatory execution requires at least 10,000 resamples, a preregistered non-inferiority margin, and Holm correction Efron and Tibshirani [1993]; Holm [1979].

3.4 Environment and reproducibility

The principal verification and integrated benchmark reruns occurred on August 24, 2026 in America/Sao_Paulo, producing UTC artifacts on August 25. Their benchmark manifest records release revision a81c566 (tag v1.4.8) plus a dirty worktree, which remains a limitation. Hardware was a Raspberry Pi 4 with four ARM Cortex-A72 cores up to 1.8 GHz and 7.6 GiB RAM, running Debian aarch64 and Linux 6.12.62. The host exposed Python 3.11.2, while the integrated benchmark environment recorded Python 3.13.13; Node.js was 20.20.2 and Docker was 29.3.1. A historical mismatch between Python package metadata and module constants was corrected to 1.4.8 and retested.

The implementation uses Python Python Software Foundation [2026], FastAPI, Pydantic, and SQLAlchemy in the API Ramírez [2026]; Pydantic Services Inc. [2026]; Bayer [2026]; Next.js and React in the frontend Vercel [2026]; Meta Open Source [2026]; and Docker for orchestration Docker Inc. [2026]. Testing and analysis use pytest, Playwright, Ruff, and uv pytest development team [2026]; Microsoft [2026]; Astral Software [2026a,b]. These citations point to primary software documentation; resolved versions and lockfiles remain the reproducibility authority.

Table 2. Software stack observed during evaluation.

Layer	Tools and versions	Role
API	Python 3.11.2; FastAPI 0.140.0; SQLAlchemy 2.0.51; Pydantic 2.13.4	HTTP, domain logic, validation, and persistence
Frontend	Next.js 15.5.22; React 19.2.7	web interface and rendering
End-to-end	Playwright 1.61.1 / Chromium	workflows, accessibility, and performance
Packaging	Docker 29.3.1	service and build isolation
Build and quality	uv 0.5.9; pytest; Ruff; MyPy	dependencies, tests, and static analysis

Table 3. Executed evidence sources.

Source	Scale	Purpose
Semantic fixture	100 notes, 45 queries	retrieval, negative queries, and latency
Retrieval ablation	44 queries	20 multi-hop, 20 factual, and 4 negative
Cognitive fixture	6 notes	extraction, connections, insights, provenance, and lifecycle
BEIR SciFact	5,183 docs, 300 queries	public independent BM25 reference
Synthetic graph	5k/20k and 10k nodes	backend, payload, memory, and rendering
Worker queue	100 jobs	enqueue, drain, claim, and duplication
Local HTTP	100 requests, concurrency 10	throughput and latency percentiles
Browser	5 repetitions per route/device	LCP, CLS, long tasks, and transfer

3.5 Ethics, privacy, and AI use

No participant data or personal notes were collected. Fixtures are not loaded during production startup. The project separates canonical vault content, derived indexes, and telemetry; exports redact secrets. A future user study must be submitted to the applicable research ethics committee and include consent, data minimization, retention, and withdrawal procedures.

Generative AI tools assisted software engineering, code review and debugging, the design and addition of automated tests, research organization, manuscript structure, and language revision. They did not autonomously establish the reported evidence or conclusions. The first author executed the test suites, inspected the retained outputs, and verified metrics, code, sources, and claims. Both authors retain full responsibility for the manuscript. No AI system is listed as an author. This disclosure follows the transparency and accountability requirements of the Brazilian Computer Society publication policy.

caches are derived and rebuildable. The runtime comprises a Next.js frontend, a FastAPI service, an asynchronous Worker, SQLite, an optional HippoRAG sidecar, and one active local or cloud model route. Local-first therefore describes ownership and recoverability; it does not imply that every inference executes locally.

4 The BerryBrain Artifact

4.1 Architecture

BerryBrain is local-first: Markdown notes remain the user-controlled canonical content, while stable identities, provenance, feedback policy, settings, and job state form authoritative operational control data. Generated graph records, lexical and vector indexes, the HippoRAG projection, and

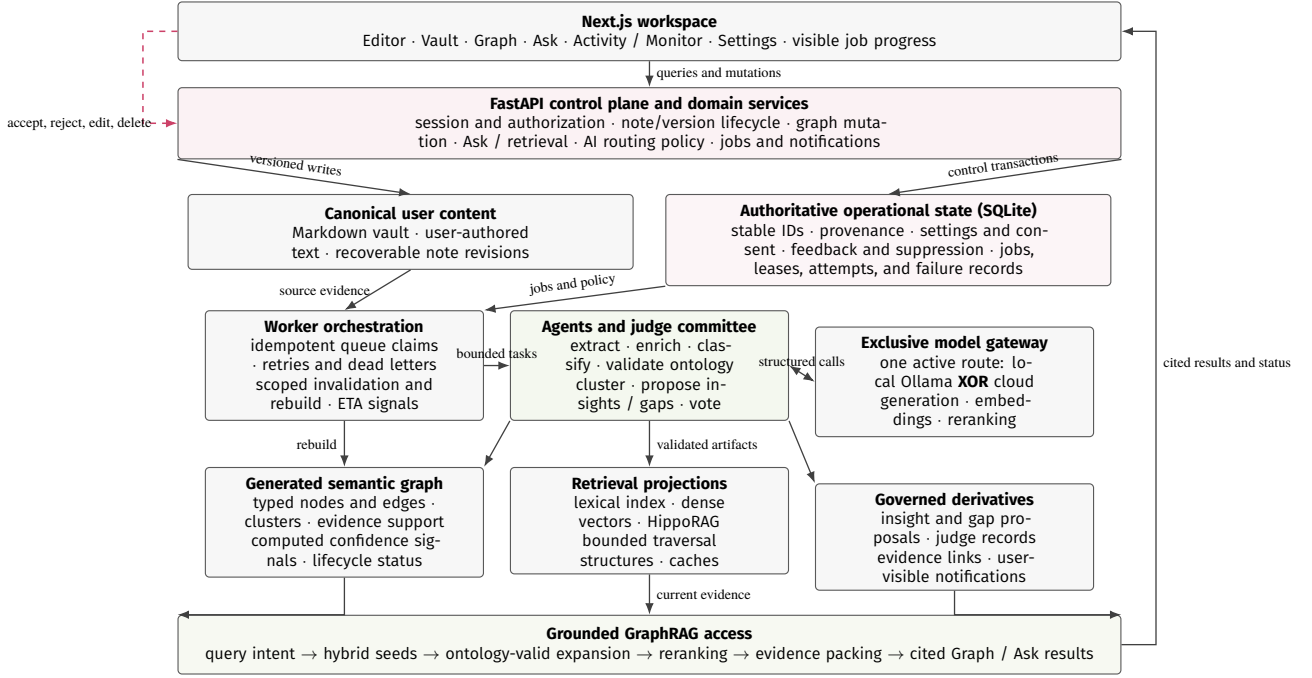


Figure 1. BerryBrain runtime, authority, and adaptation boundaries. Solid arrows show transactional, job, model, artifact, or retrieval flow; the dashed arrow shows auditable user feedback and lifecycle actions. Markdown is canonical user content, SQLite retains authoritative control state, and graph, retrieval, and proposal structures are derived and rebuildable.

Cloud providers and Ollama are mutually selected runtime modes. The active configuration determines the route; local calls must not occur while cloud mode is active. Jobs expose status, attempts, failure reasons, and operational estimates. Autopilot schedules enrichment, insights, gaps, judge evaluation, and clustering without requiring manual feature buttons.

4.2 Operational ontology

The ontology prevents the graph from becoming a visualization of arbitrary words. Each class has a meaning, visual shape, and naming rules. Each predicate is a directed verb phrase with domain, range, symmetry, and admissible evidence. Navigation fragments, isolated verbs, stop phrases, unsupported abbreviations, and diagnostics are rejected or quarantined.

Table 4. Canonical node classes.

Class	Shape	Semantics
Note	circle	canonical document and provenance root
Concept	diamond	reusable abstract idea expressed by sources
Entity	hexagon	identifiable person, organization, place, product, event, or work
Topic	triangle	broad thematic group for navigation and retrieval
Insight	rectangle	evidence-supported proposal subject to acceptance
ResearchGap	octagon	unresolved question or missing evidence

Shape encodes class; color encodes semantic cluster; color never encodes confidence. Every current cluster receives a deterministic contrast-safe color. A parent note uses the primary #CC4168 color, and insights receive a highlight while remaining rectangular. Canonical predicates include mentions, defines, supports, contradicts, broaderThan, narrowerThan,

relatedTo, derivedFrom, and evidencedBy. The machine-readable contract validates permitted endpoints before admission.

4.3 Retrieval and grounded answers

Ask accepts questions about graph content and graph topology. The pipeline retrieves lexical and dense seeds, applies fusion, expands only ontology-valid paths, removes stale or suppressed sources, reranks with conservative support, and passes cited evidence to the generator. If a provider fails, the interface does not display an answer-like fallback. It preserves a waiting or error state and offers retry and configuration actions. Question suggestions are generated only when relevant graph content exists.

4.4 Signal-derived evidence-support interval

Clients cannot write confidence. Given distinct normalized signals $x_i \in [0, 1]$, the runtime computes their mean $\bar{x}$ and a nominal two-sided, empirical-Bernstein-style bounded interval. The implementation follows the radius form of Audibert et al. Audibert *et al.* [2009], uses the Bessel-corrected sample variance s^2 , and fixes the nominal level at $1 - \delta = 0.95$. For $n > 1$:

$$r = \sqrt{\frac{2s^2 \ln(3/\delta)}{n}} + \frac{3 \ln(3/\delta)}{n}, \quad (1)$$

$$L = \max(0, \bar{x} - r),$$

$$U = \min(1, \bar{x} + r).$$

Bounds are clipped to $[0, 1]$. The displayed score is the midpoint $(L + U)/2$ of those clipped bounds; downstream ranking may use L conservatively. One scored

signal deliberately yields $[0, 1]$, and no scored observations yield an unavailable state. Provenance identifiers remain listed as factors but do not increase n , avoiding one form of double-counting. The serialized method is `empirical-bernstein-bounded-signals-v1`.

The theoretical concentration guarantee assumes bounded independent observations estimating a common expectation. BerryBrain combines heterogeneous heuristic, model, and user signals whose sampling process has not been validated against those assumptions. Consequently, “95%” is a nominal construction level and the output is an *evidence-support uncertainty interval*, not a frequentist coverage claim for factual truth and not a calibrated correctness probability. This runtime interval is also distinct from the paired percentile bootstrap used for retrieval effects.

4.5 Feedback and lifecycle

Semantically meaningful actions generate append-only learning events. Creating or materially editing a note schedules a new version and detaches obsolete provenance when required. Renaming or moving preserves identity. Deletion removes note-owned evidence and recalculates only dependent neighborhoods. Confirming, ignoring, correcting, restoring, or deleting a node or edge updates an active policy by canonical identity and source scope. Accepting or rejecting an insight and correcting an Ask answer also produce policy signals. Navigation and incidental clicks are not learning labels.

Before deleting a derived node, the API captures incident edges, sources, neighbors, cluster, and dependent insights. Within the transaction, it expires insights, quarantines relationships that cite the artifact, removes projections, and records negative feedback. It then schedules scoped relationship reconciliation, insight rediscovery, judging, and reclustering. Consequently, a rejected identity must not silently reappear after refresh, and relationships that depended on it must not remain active.

4.6 Agents and judge committee

Agents propose extraction, enrichment, connections, clusters, insights, gaps, and answers; they do not directly admit knowledge. Deterministic validators enforce type, naming, provenance, state, and ontology rules. Judges assess faithfulness, relevance, contradiction, source quality, and ontology consistency.

The committee discovers the live provider catalog and sends structured probes through the same runtime route. Embedding, reranking, moderation, audio, image, and active generator models are excluded. Each slot persists provider, model, role, and focus and remains editable in Settings. The committee requires at least two distinct valid verdicts. Failed calls remain recorded but are excluded from voting. Catalog presence is insufficient because listed models may still return 404, time out, or reject the required schema.

5 Experimental Protocol

5.1 Software verification

The evaluation executed unit, integration, subtest, Chromium E2E, lint, typecheck, production build, architecture fitness, security, dependency vulnerability, MyPy, and coverage checks. Overlapping test counts are not summed as independent observations.

5.2 Retrieval ablation

The fixture includes multi-hop queries intentionally requiring graph paths, factual queries, and negative queries. Configurations are lexical, dense, standard hybrid, graph lexical, and graph-hybrid. Because this is a designed regression test, perfect scores only show that the implementation recovers expected paths.

5.3 External benchmark

The official BEIR `scifact.zip` archive was downloaded with SHA-256 `536e14446a0ba56ed1398ab1055f39fe852686ecad24a6306c80c490fa8e0165`. An independent BM25 implementation indexed 5,183 documents and evaluated 300 test queries. This run verifies the baseline and benchmark infrastructure on public data. The complete BerryBrain pipeline did not run on the same corpus in this stage, so the numbers do not support a direct superiority claim. The retained external bundle was generated on August 14 at revision `c297558`; it is reported separately rather than represented as part of the August 25 integrated run.

5.4 Performance, fault, and frontend tests

Integrated profile S exercised 5,000 nodes and 20,000 edges over seven samples. An on-disk benchmark used 500 nodes and 1,000 edges. The queue processed 100 jobs. The HTTP test sent 100 requests at concurrency 10. Fault injection simulated interruption, invalid jobs, and scoped recomputation while checking containment and state preservation.

Playwright measured public routes on desktop and emulated mobile with five repetitions per cell. A separate scenario measured 10,000 graph nodes, first visual, complete load, p95 interaction, and heap. Automated WCAG 2.2 A/AA checks were included, although automation does not replace manual assessment W3C [2023].

Table 5. Executed software verification suites.

Suite	Passed	Failed	Note
API functional	465	0	59 subtests
API under coverage	464	0	1 expected timing skip
Worker	55	0	integrated API import path
Chromium E2E	56	0	4.1 minutes
Focused after fixes	33	0	benchmark and judge
Ruff / MyPy	all	0	6 files in MyPy scope
Web lint/typecheck/build	all	0	23 routes, 103 kB shared bundle
Architecture fitness	12 checks	0	pass
Security audit	9 checks	0	pass

6 Results

6.1 Retrieval quality

Table 6. Internal ablation over 44 queries (R@10 denotes Recall@10).

Method	R@10	MRR	nDCG@10	p50 ms	p95 ms
Lexical	0.05	0.0167	0.0247	7.75	9.73
Dense	0.50	0.50	0.50	7.27	8.90
Standard hybrid	0.50	0.50	0.50	14.94	16.69
Graph lexical	0.50	0.25	0.3155	15.10	18.16
Graph-hybrid	1.00	0.75	0.8155	21.54	25.80

On 20 multi-hop queries, standard hybrid Recall@10 was 0 and graph-hybrid was 1.00; the difference was 1.00 with an exploratory paired bootstrap 95% CI of [1.00, 1.00]. On 20 factual queries, both scored 1.00. Citation precision, evidence faithfulness, negative rejection, and stale evidence removal were 1.00 in the fixture. Graph-hybrid added 9.11 ms at p95 over standard hybrid, consistent with expansion and reranking.

The separate semantic fixture contained 100 notes and 45 queries. It achieved Recall@5 of 0.50, Recall@10 of 1.00, MRR and nDCG@10 of 1.00, negative rejection of 1.00, and p95 latency of 45.85 ms. Perfect values reflect deterministic queries and do not establish generalization.

Table 7. Independent BM25 result on BEIR SciFact.

Metric	Result	Scale
Recall@10	0.781611	300 queries
MRR	0.638610	300 queries
nDCG@10	0.664390	300 queries
Index construction	1,393.12 ms	5,183 documents
p50 / p95 / p99 latency	309.78 / 386.21 / 406.30 ms	per query

6.2 Cognitive quality, lifecycle, and judges

On the six-note cognitive fixture, concept and connection precision and recall were 1.00; grounded insight and provenance

coverage were 1.00; unsupported claims, diagnostic leakage, and stale artifact rates were zero. Idempotent rebuild, stale removal, and review preservation were true. Across 12 insight fixtures, precision, recall, accuracy, and usefulness were 1.00. These are regression gates, not human semantic evaluation.

The judge ran 100 evaluations and 30 synthetic reference reviews, with 29 matches, weighted kappa of 0.9801, and zero false acceptances or rejections in that set. No human review was collected. The correct status is therefore `regression_only` and `calibrated=false`; no human agreement or committee superiority is concluded.

Table 8. Cognitive and judge evidence.

Aspect	Result	Permitted interpretation
Concept / connection	P=1.00; R=1.00	fixture contract satisfied
Insights	P=1.00; R=1.00	expected proposals recovered
Provenance	1.00	every fixture derivative has a source
Stale residue	0	invalidation worked in the scenario
Synthetic judge	$\kappa_w = 0.9801$	regression stability
Human calibration	absent	external validity not established

6.3 Coverage, static quality, and security

Table 9. Statement coverage.

Component	Statements	Covered	Missing	Coverage
API + benchmarks	16,334	13,425	2,909	82.19%
Worker	1,416	693	723	48.94%

The largest coverage risks are external gateways and integration paths: semantic graph service (16.44%), HippoRAG router (27%), vault-to-graph pipeline (28.57%), AI configuration (38.74%), Worker Ollama gateway (15%), and cloud gateway (16.96%). High aggregate API coverage therefore does not imply complete validation of live providers.

The frontend audit found zero vulnerabilities across 160 production dependencies; the API found zero across 63 dependencies. A stale Worker development virtual environment contained pypdf 6.14.2, affected by two 2026 advisories, but pypdf is absent from the Worker production man-

ifest, and the API requires version 6.15.0 or newer. This is environment drift, not a deployed vulnerability. The project security audit passed 9/9 controls, including no tracked vault content, no hardcoded secrets, protected keys, redacted exports, and opt-in external research.

6.4 Backend and queue performance

Table 10. Backend results on Raspberry Pi 4.

Scenario	p50	p95	Additional result
Graph 5k/20k	2,687.21 ms	2,739.55 ms	14.15 MB; 63.96 MB peak
On-disk graph 500/1k	183.97 ms	201.07 ms	864 kB; 3.93 MB peak
HTTP, 100 requests	154.29 ms	274.36 ms	62.02 req/s; 0 failures
Queue, 100 jobs	6,021.58 ms	8,736.56 ms	p99 8,997.23 ms
Job claim	–	59.69 ms	0 duplicates
Injected faults	–	9.37 ms max	3/3 contained

The 5k graph benchmark met budgets of 5 s, 16 MiB payload, and 512 MiB memory. The queue enqueued 49.24 jobs/s and drained 11.03 jobs/s; all completed. The high completion p95 reflects serialized work on this hardware and should inform user-visible ETA. Fault injection preserved canonical state and recovered in under 10 ms in the simulated cases; prolonged real provider outages were not reproduced.

6.5 Frontend performance

Table 11. Public-route browser measurements (milliseconds except CLS; five repetitions per cell).

Profile	Route	p50	p95	LCP p75	CLS p75
Desktop	/	1,815.16	2,383.16	596	0.00130
Desktop	/docs	1,402.57	1,574.19	748	0
Desktop	/faq	1,271.30	1,572.63	440	0.00037
Mobile	/	1,654.57	1,907.27	360	0.00096
Mobile	/docs	1,245.69	1,267.05	456	0
Mobile	/faq	1,206.66	1,537.90	336	0.00096

LCP and CLS remained within commonly accepted good thresholds in this local lab run. Maximum p95 long-task duration was 782.2 ms on desktop (/docs) and 553.2 ms on mobile (/), indicating main-thread work to reduce. All 30 route/profile observations recorded zero page and HTTP 5xx errors. Transfer for /docs was recorded as zero because the browser cache was warm and was therefore excluded from network inference.

For the synthetic 10,000-node/40,000-edge browser graph, cold first visual was 2,186.32 ms, warm first visual was 1,018.29 ms, complete load was 6,349.92 ms, interaction p95 was 33.5 ms, and used heap was 23.1 MB. The view remained interactive after load, but the 6.35 s completion time still requires progressive loading, cache, and explicit progress feedback.

6.6 Maturity

The release gate passed with no failed mandatory gates. Median maturity was 2 and minimum maturity was zero. Nine groups reached level 2; capture/extraction and security remained at level 0 because the maturity registry lacked cur-

rent evidence entries for those groups. This does not contradict the separate 9/9 security audit: the instruments consume different evidence registries. The overall classification is incomplete-evidence, appropriate for a functional research prototype rather than a product validated in longitudinal use.

7 Discussion

7.1 Answers to the research questions

Table 12. Answers supported by the current evidence.

RQ	Status	Answer
RQ1	exploratory support	graph-hybrid recovered fixture multi-hop paths with no observed factual loss
RQ2	regression support	provenance, negatives, and stale deletion passed fixtures
RQ3	regression support	idempotence and lifecycle passed; no real corpus was annotated
RQ4	partial	support is signal-derived and auditable; nominal coverage and correctness calibration are unvalidated
RQ5	not confirmed	synthetic kappa is high, but there were zero human reviews
RQ6	locally answered	costs were measured on Pi 4; synthetic 10k graph load took 6.35 s
RQ7	unanswered	no participant study was executed
RQ8	mechanism validated	events and suppression pass tests; longitudinal effect is unknown
RQ9	partial support	3/3 simulated faults were contained; prolonged real faults remain open

The main result is architectural: structural expansion produces an extreme gain on the set designed to require relationships. This demonstrates that the feature exists and is measurable, but the degenerate [1.00, 1.00] interval does not represent real-world variability. A superiority claim requires BerryBrain, BM25, DPR/hybrid, vanilla RAG, and graph methods on the same public or annotated corpus, under identical qrels and budgets.

Graph-hybrid query cost was small in the fixture, but large-graph materialization cost was visible. The appropriate design is not to choose quality or speed globally. Expansion should be bounded, cached, and conditioned on query intent. Simple factual queries may stop early; global or multi-hop questions justify graph traversal.

7.2 Ontological meaning and feedback learning

An ontology is useful when it constrains operations, not when it merely names shapes. In *BerryBrain*, class and predicate affect validation, expansion, inverse relationships, evidence, visualization, and lifecycle. A claim that two unrelated notes are connected because both contain an isolated word such as *skip* must be rejected when the token is a navigation fragment, isolated verb, or contextually unsupported. If the user deletes that concept, negative policy, scope, and dependent invalidation prevent the same identity from being silently readmitted.

This behavior is non-parametric feedback learning: it updates external memory and policy rather than model weights Gutiérrez *et al.* [2025]. It may improve later decisions in the same context, but it can also over-suppress. Decisions are therefore versioned, reversible, and scoped; a local rejection

must not be generalized to the entire vault without evidence. Longitudinal evaluation must measure rejected-artifact recurrence and false suppression over time.

7.3 A valid comparative strategy

Three comparisons are complementary:

1. **BerryBrain against itself:** ablation isolates graph, judge, enrichment, and feedback;
2. **independent implementations on the same task:** BM25, dense, hybrid, vanilla RAG, and GraphRAG share corpus, qrels, context, and model budgets;
3. **human workflow:** conventional search, vector RAG, and BerryBrain are compared by success, time, evidence, SUS, and NASA-TLX Brooke [1996]; Hart and Staveland [1988].

Comparing the internal 44-query result directly with SciFact would be invalid because corpus, qrels, and difficulty differ. The external table contextualizes a baseline without mixing populations.

8 Threats to Validity and Limitations

Construct validity. Fixtures measure known contracts and may favor the implementation. Perfect scores have low resolution. The synthetic judge does not measure human trust. System confidence is not a calibrated probability.

Internal validity. The worktree was modified, and the exploratory profile used only 2,000 resamples. Cache, temperature, concurrent services, and nearly exhausted swap may affect latency. Test suites overlap, so counts are not independent samples.

External validity. Execution used one Raspberry Pi, no personal vaults, and no varied language or provider population. The public corpus was used only for BM25. Cloud-provider outages and multiple live LLM families were not evaluated.

Conclusion validity. Twenty multi-hop queries are insufficient to estimate a real-world effect. No power analysis, blinded annotation, confirmatory non-inferiority test, or multiple-test correction was applied to this exploratory result.

Software coverage. Worker and external gateway branches remain under-tested. Automated WCAG E2E tests do not replace keyboard, screen-reader, manual contrast, and user evaluation.

Publication readiness. This manuscript is written in English and uses the JBCS template, as required by the journal. The first author's ORCID was resolved through the official public registry and is included in the author block. The second author's ORCID must be supplied and verified before submission. The CRediT statement, target area, and submission metadata are recorded with the artifact. The immutable GitHub research release is identified below; its DOI-bearing Zenodo deposit remains pending and is not claimed. Independent language review, both authors' approval, and explicit acceptance of the journal's reviewer commitment remain pre-submission actions that cannot be delegated to an automated system.

9 Confirmatory Evaluation Plan

The confirmatory study should freeze commit, dependencies, prompts, and providers; execute BM25, dense, hybrid, vanilla RAG, GraphRAG without judges, and full *BerryBrain* on the same corpus; cover factual, multi-hop, global, negative, contradictory, and temporal queries; and publish qrels and redacted logs. Two independent annotators should label nodes, edges, citations, and answers, with adjudication and inter-rater agreement.

Each model judge verdict should be compared with blinded human reviewers, reporting weighted kappa, critical false acceptance, false rejection, and position bias. Confidence should be evaluated with Brier score, expected calibration error, reliability diagrams, and interval coverage Brier [1950]; Guo *et al.* [2017].

The user study should use a counterbalanced design, equivalent tasks, and a power-based sample size. Outcomes include success, time, correct evidence, SUS, NASA-TLX, predicted confidence, and observed correctness. A longitudinal phase should measure recurrence after rejection, correction retention, false suppression, and reversibility. None of these result fields should be populated before execution.

10 Conclusion

This work presented *BerryBrain*, an adaptive architecture that transforms notes into a governed semantic graph and queries the collection through GraphRAG with provenance, signal-derived confidence, configurable judges, and human feedback. Reproducible evaluation passed the principal software suites, showed a multi-hop gain on the designed fixture, preserved factual retrieval in that set, invalidated stale artifacts in exercised scenarios, and quantified performance on constrained hardware.

The data also bound the maturity claim. The system is a functional research prototype with good API coverage, moderate Worker coverage, long tasks on public routes, a 6.35 s complete synthetic 10,000-node graph load, and gaps in live external integrations. There is no evidence yet of human calibration, real-task advantage, or longitudinal adaptation. The most defensible contribution is therefore the auditable artifact and its evaluation protocol, not a universal superiority claim.

Authors' Contributions

Mateus Almeida de Souza: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, and Writing – review & editing. Leonardo José Silvestre: Supervision and Writing – review & editing.

Competing interests

The authors declare no known competing interests related to this work.

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Availability of data and materials

Source code, benchmark scripts, synthetic fixtures, protocols, and retained evidence are preserved in the BerryBrain JBCS research-artifact release. The release records the source snapshot, lock-files, hashes, non-secret configuration, and reproduction instructions. Personal vault data and secrets are not distributed. A DOI-bearing Zenodo deposit remains a pre-submission requirement; it has not yet been created and no DOI is claimed by this manuscript.

References

- Astral Software (2026a). Ruff documentation. Software documentation.
- Astral Software (2026b). uv documentation. Software documentation.
- Audibert, J.-Y., Munos, R., and Szepesvári, C. (2009). Exploration–exploitation tradeoff using variance estimates in multi-armed bandits. *Theoretical Computer Science*, 410(19):1876–1902. DOI: 10.1016/j.tcs.2009.01.016.
- Balog, K. and Kenter, T. (2019). Personal knowledge graphs. In *Proceedings of the 2019 ACM SIGIR International Conference on Theory of Information Retrieval*, pages 217–220. DOI: 10.1145/3341981.3344241.
- Bayer, M. (2026). Sqlalchemy documentation. Software documentation.
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3. DOI: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- Brooke, J. (1996). Sus: A quick and dirty usability scale. In Jordan, P. W., Thomas, B., McClelland, I. L., and Weerdmeester, B., editors, *Usability Evaluation in Industry*, pages 189–194. Taylor and Francis. DOI: 10.1201/9781498710411-35.
- Chakraborty, P. and Sanyal, D. K. (2023). A comprehensive survey of personal knowledge graphs. *WIREs Data Mining and Knowledge Discovery*, 13(5):e1513. DOI: 10.1002/widm.1513.
- Cormack, G. V., Clarke, C. L. A., and Buettcher, S. (2009). Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. DOI: 10.1145/1571941.1572114.
- Docker Inc. (2026). Docker documentation. Software documentation.
- Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., and Larson, J. (2024). From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*. DOI: 10.48550/arXiv.2404.16130.
- Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman and Hall/CRC.
- Es, S., James, J., Espinosa Anke, L., and Schockaert, S. (2024). Ragas: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158. DOI: 10.18653/v1/2024.eacl-demo.16.
- Fraga, F., Poggi, M., Casanova, M., and Leme, L. (2023). On the automatic generation of knowledge connections. In *Proceedings of the 25th International Conference on Enterprise Information Systems*, pages 348–355. DOI: 10.5220/0011781100003467.
- Fraga, F. P. A., Poggi, M., Casanova, M. A., and Paes Leme, L. A. P. (2024). Creating automatic connections for personal knowledge management. *SN Computer Science*, 5:737. DOI: 10.1007/s42979-024-02876-4.
- Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pages 1321–1330. DOI: 10.48550/arXiv.1706.04599.
- Guo, Z., Xia, L., Yu, Y., Ao, T., and Huang, C. (2024). Lightrag: Simple and fast retrieval-augmented generation. *arXiv preprint arXiv:2410.05779*. DOI: 10.48550/arXiv.2410.05779.
- Gutiérrez, B., Shu, Y., Gu, Y., Yasunaga, M., and Su, Y. (2024). Hipporag: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 11998–12032. DOI: 10.52202/079017-1902.
- Gutiérrez, B. J., Shu, Y., Qi, W., Zhou, S., and Su, Y. (2025). From rag to memory: Non-parametric continual learning for large language models. In Singh, A., Fazel, M., Hsu, D., Lacoste-Julien, S., Berkenkamp, F., Maharaj, T., Wagstaff, K., and Zhu, J., editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 21497–21515. PMLR. DOI: 10.48550/arXiv.2502.14802.
- Hart, S. G. and Staveland, L. E. (1988). Development of nasa-tlx: Results of empirical and theoretical research. In Hancock, P. A. and Meshkati, N., editors, *Human Mental Workload*, pages 139–183. North-Holland. DOI: 10.1016/S0166-4115(08)62386-9.
- Hevner, A. R., March, S. T., Park, J., and Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1):75–105. DOI: 10.2307/25148625.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., and Yih, W.-t. (2020). Dense passage

- retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781. DOI: 10.18653/v1/2020.emnlp-main.550.
- Kim, S., Suk, J., Longpre, S., Lin, B. Y., Shin, J., Welleck, S., Neubig, G., Lee, M., Lee, K., and Seo, M. (2024). Prometheus 2: An open source language model specialized in evaluating other language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 4334–4353. DOI: 10.18653/v1/2024.emnlp-main.248.
- Lebo, T., Sahoo, S., and McGuinness, D. (2013). Prov-o: The prov ontology. W3C Recommendation.
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. DOI: 10.48550/arXiv.2005.11401.
- Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., and Zhu, C. (2023). G-eval: Nlg evaluation using gpt-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522. DOI: 10.18653/v1/2023.emnlp-main.153.
- Meta Open Source (2026). React documentation. Software documentation.
- Microsoft (2026). Playwright documentation. Software documentation.
- Peffers, K., Tuunanen, T., Rothenberger, M. A., and Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3):45–77. DOI: 10.2753/MIS0742-1222240302.
- Pydantic Services Inc. (2026). Pydantic documentation. Software documentation.
- pytest development team (2026). pytest documentation. Software documentation.
- Python Software Foundation (2026). Python 3.11 documentation. Software documentation.
- Ramírez, S. (2026). Fastapi documentation. Software documentation.
- Saad-Falcon, J., Khattab, O., Potts, C., and Zaharia, M. (2024). Ares: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 338–354. DOI: 10.18653/v1/2024.naacl-long.20.
- Shi, L., Ma, C., Liang, W., Diao, X., Ma, W., and Vosoughi, S. (2025). Judging the judges: A systematic study of position bias in llm-as-a-judge. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 304–322. DOI: 10.18653/v1/2025.ijcnlp-long.18.
- Skjæveland, M. G., Balog, K., Bernard, N., Łajewska, W., and Linjordet, T. (2024). An ecosystem for personal knowledge graphs: A survey and research roadmap. *AI Open*, 5:55–69. DOI: 10.1016/j.aiopen.2024.01.003.
- Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., and Gurevych, I. (2021). Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1. DOI: 10.48550/arXiv.2104.08663.
- Trivedi, H., Balasubramanian, N., Khot, T., and Sabharwal, A. (2022). Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554. DOI: 10.1162/tacl_a_00475.
- Vercel (2026). Next.js documentation. Software documentation.
- W3C (2009). Skos simple knowledge organization system reference. W3C Recommendation.
- W3C (2012). Owl 2 web ontology language document overview. W3C Recommendation.
- W3C (2014). Rdf 1.1 concepts and abstract syntax. W3C Recommendation.
- W3C (2023). Web content accessibility guidelines (wcag) 2.2. W3C Recommendation.
- Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W., Salakhutdinov, R., and Manning, C. D. (2018). Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380. DOI: 10.18653/v1/D18-1259.